Numerical Algorithms for Physics: Project List

Andrea Mignone
Physics Department, University of Torino
AA 2025-2026

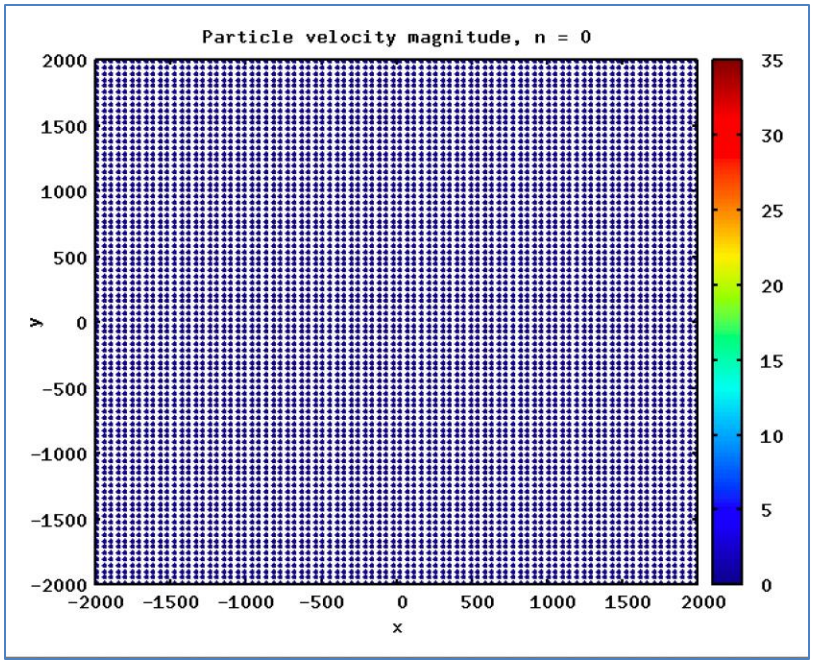
Writing the Project

- In the following I will show a list of projects for this course.
- Each project should be no more than ≈ 10-20 pages long (excluding the code listed in the appendix).
- A .pdf file is strongly recommended.
- The document structure should consists of
  - An abstract / Introductory part where the physical problem is explained and why we need to resort to numerical integration.
  - A test section where the numerical method is validated against known analytical / reference solution.
  - A model study of the problem including plots.
  - A final summary/discussion.
  - An appendix including the code used for the project, using fixed-size fonts (a font whose letters and characters each occupy the same amount of horizontal space, e.g., Courier, Courier New, Lucida Console, Monaco, and Consolas).

Project #7: Particle(s) in EM Fields

- Study the motion of one or more (non-relativistic) particles in a fixed electromagnetic field:

$$\frac{d\vec{v}}{dt} = \frac{e}{m}\left(\vec{E} + \frac{\vec{v}}{c} \times \vec{B}\right)$$



- The previous equation is written in the c.g.s sytem (widely used in astrophysics) but dimensionless units are recommended.

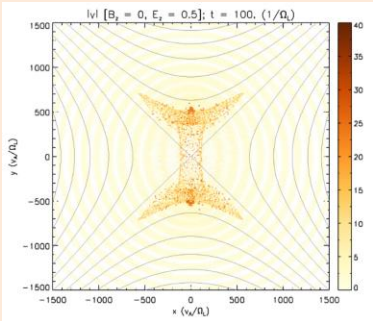
- The equation of motion (possibly) involves propagation in all 3 direction (x,y,z).

- The electric and magnetic field vector are given externally and particles do not interact with each other.

- The project involves direct integration of the equation of motion using RK-type integrators and/or the generalization of symplectic integration scheme to the case of velocity-dependent force (the Boris algorithm is the progenitor of such schemes).

Project #7: Particle(s) in EM Fields

- Provide a project with at least three different cases. A suggestion is given here:

Case	***N_{part}***	***E***	***B***	***Notes***
Simple gyration	1	(0,0,0)	(0,0,1)	Consider both perpendicular and parallel propagation. Check your results with analytical formula.
ExB drift	1	(0,E,0) with E < 1	(0,0,1)	Consider propagation along y-direction: what motion do you see ? Does the particle accelerate ? Explain.
X point  	10^3	(0,0,1/2)	(y/L, x/L,0) L=10^3.	Place particles uniformly in the square domain $[-L,L]^2$ and initialize particle velocity to 0.1 using randomly numbers distributed angles. Describe your results: - What kind of motion is observed ? - Do particles accelerate ? Where ?